# PETROLEUM PRODUCTION ENGINEERING

# MODULE-11

## INTRODUCTION TO ARTIFICIAL LIFT TECHNIQUES

## SUCKER ROD PUMP

By
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## Sucker Rod Pump

- Sucker Rod Pump is the oldest and most widely used type of artificial lift.
- Sucker rod pump is also referred to as "*Beam Pumping Unit*" or "*Nodding Donkey*".
- It provides mechanical energy to lift oil from the bottom hole to the surface.
- The whole pumping system can be classified under three broad units, namely:
  - Surface Unit
  - Sucker Rods
  - Sub Surface Pump

## SRP-Surface Unit

- It comprises of the pumping unit cum prime mover which converts the rotary motion into the reciprocating/vertical motion with the help of several link arrangements.
- Majority of this pumping operation utilise walking beam pumping unit and so it is named as beam pumping unit.
- The pumping units are broadly classified into two groups:
  - *Conventional unit* where walking beam operates as a double arm Class-I lever
  - *Non-conventional unit* where walking beam operates as a single arm Class-III lever.

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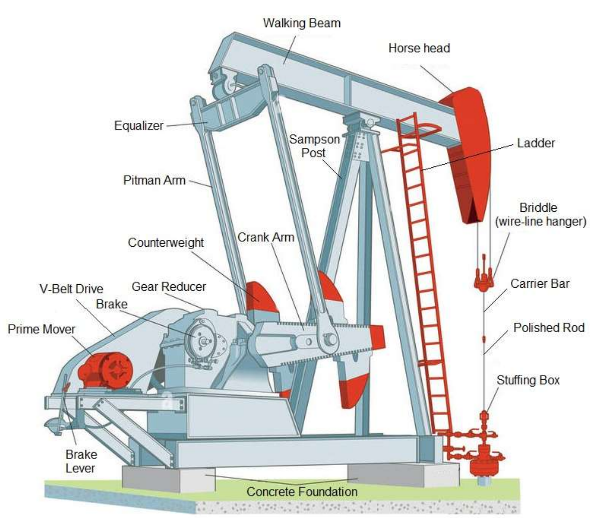
## SRP-Surface Unit

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## Components of Surface Units

- ***Prime Mover***: The two commonly used types of prime movers are either electric motors or internal combustion engines that run on gas
- ***V-belt Drive***: It transmits the rotary motion from the prime mover to the gear reducer shaft.
- ***Crank Arm***: These are connected to the crankshaft of the gear reducer assembly. Each of the arms has three or four holes for attaching the pitman arms, depending on the desired stroke length.
- ***Pitman Arm***: A connecting rod that links that is connected to the crank at one end and the rear of the walking beam at the other end and converts the crank's rotary motion into a reciprocating, up/down motion.



## Components of Surface Units

- ***Walking Beam***: It transmits energy from the prime mover to the sucker rod string
- ***Sampson Post***: It consists of three or four legs of rolled steel to support the walking beam, horsehead, equalizer, pitman arms, and more than twice the peak sucker rod load.
- ***Horse Head***: It (Mulehead) is attached to one end of the walking beam and is positioned so that the polished rod is directly over the wellhead.
- ***Equaliser***: The equalizer connects the pitman arm to the walking beam with an equalizer bearing (tail bearing).
- ***Briddle (Wire-Line Hanger)***: Bridle hangs vertically from the horsehead and is fastened to the carrier bar.

## Components of Surface Units

- *Counterweight*: The counterweights are attached to the crank arms and rotate with the crankshaft. During the upstroke, the counterweights rotate downwards, reducing the amount of energy needed to lift fluid. During downstroke, the weight of the dropping sucker rod string provides the energy to lift the counterweights and store the energy in the process.
- *Double reduction gearbox*: The gear reducer (speed reducer) has gears to convert the prime mover's high-speed, low-torque output to the low-speed, high-torque output required by the pumping unit. Most conventional pumping units use double reduction gear trains, which reduce speed in two stages.
- *Ladder*: It is attached to the Sampson post for easy and safe accessibility for repair/maintenance of the walking beam, horsehead, and various bearings.



## Components of Surface Units

- *Carrier Bar*: As the horsehead moves up and down in a curved path, the carrier bar moves in a perfectly vertical line, directly over the wellhead restricting its wobbling.
- *Brake*: It is designed to restrain motion in all rotary joints. It is often composed of a disk or drum mounted on the reducer input shaft combined with a mechanism to impart a restraining friction torque.
- *Polished Rod*: It passes through the carrier bar and is secured by a polished rod clamp
- *Stuffing Box*: In SRP wells, a stuffing box replaces the crown valve. The packing is made of rubber elements with metallic supports.

## SRP-Sucker Rods

- The sucker rod or sucker rod string is the most crucial link between the sub-surface pump and the pumping unit.
- They are solid steel bars connected up to the depth of the pump.
- Sucker rod string also includes ***Sinker Bars*** placed just above the pump to provide stability to the downhole SRP operations due to their increased weight on the pump.
- Sucker rods are available in three different lengths i.e., 25, 30, and 35 ft.
- The diameter of the rod body ranges from ½ to 1⅛ in. with ⅛ in. increments.

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## SRP-Sub Surface Pump

- It is a subsurface reciprocating pump actuated by the up and down motion of sucker rods which connect the pipe between the surface unit and the subsurface pump.
- Its main components are:
  - Barrel
  - Plunger
  - Standing valve
  - Travelling valve
  - Pump seat

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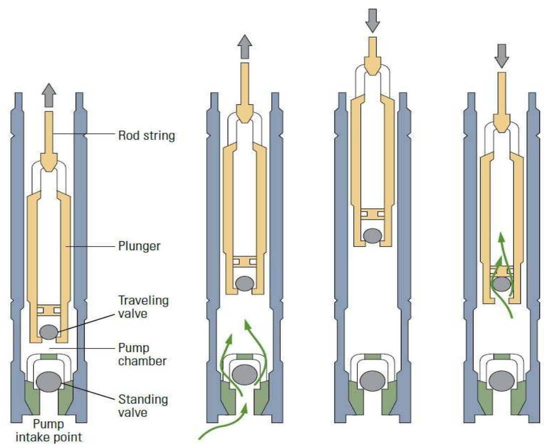
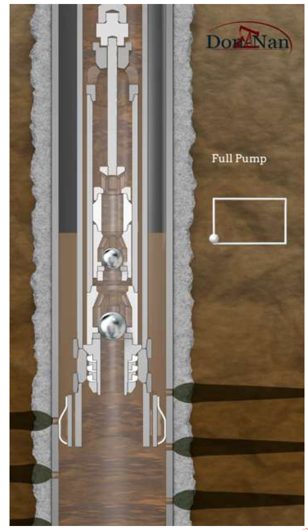
## SRP-Sub Surface Pump






## Pumping Cycle

- A conventional pump consists of a working barrel or liner suspended on the tubing; the plunger is moved up and down inside this barrel by the sucker rod string.
- At the bottom of the working barrel is a stationary ball-and-seat valve called as the standing valve while a second ball-and-seat valve called as the traveling valve is located in the plunger.
- Both the valves are unidirectional and allow fluid to pass through them in the upward direction only.

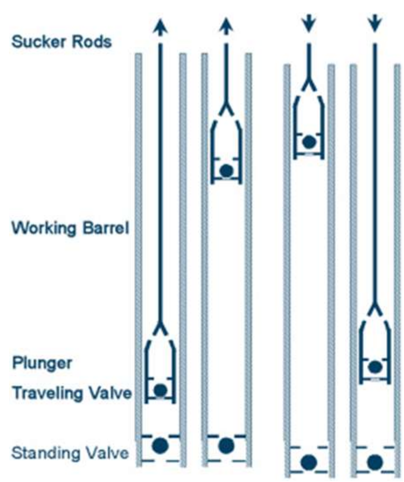
## Pumping Cycle

- As the plunger is moved downward by the sucker rod string, the traveling valve is open whereas the standing valve is closed, which forces the well fluids to enter the plunger through the traveling valve, as it moves to a position just above the standing valve.
- It consequently closes when the weight of the fluid column overcomes the bottom hole pressure.
- When the plunger is at the bottom of the stroke and is about to start an upward stroke, the traveling valve closes and the standing valve opens as suction creates pressure less than the bottom hole pressure.



## Pumping Cycle

- As upward motion continues, the fluid in the well below the standing valve is drawn into the volume above the standing valve.
- The fluid continues to fill the volume above the standing valve until the plunger reaches the top of its stroke.



# PETROLEUM PRODUCTION ENGINEERING

# MODULE-12

## INTRODUCTION TO ARTIFICIAL LIFT TECHNIQUES

## GAS LIFT

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## Gas Lift

- Gas lift is one of the methods of Artificial Lift whereby relative high-pressure gas is injected into a fluid column to reduce the density of the fluid from the point of gas injection to the surface.
- The injection gas supplements the formation gas and lightens the flowing fluid traverse above the point of injection.
- This allows a lower reservoir pressure to lift the fluids to the surface.
- The premature application of gas lift adopted the simple U-tube or pin-hole principle in the production of oil from shallow wells, later with the advent of gas lift valves, the gas lift application was extended to deeper wells.

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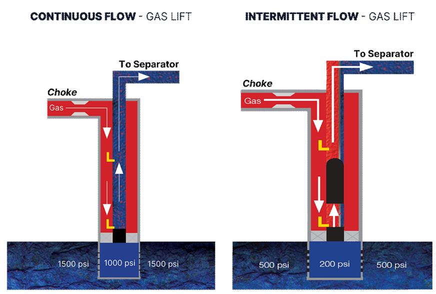
## Types of Gas Lift

- Gas lift Systems can broadly be classified into two main categories:
  1. Continuous Gas Lift
  2. Intermittent Gas Lift







## Continuous Gas Lift

- Continuous gas lift resembles the extension of self-flow wherein gas is injected in the liquid column at some predetermined depth at a controlled rate to aerate the liquid column above it and as a result the density of liquid column gets reduced to a point where a flowing bottom hole pressure for a desired rate of production is sufficient to lift the liquid to the surface.
- Thus, liquid is produced continuously from the well.
- The reservoir should be in a condition to give influx continuously.
- It should have good Bottom Hole Flowing Pressure and Productivity Index.
- Continuous gas lift characteristically provides high volume oil production having good efficiency.

## Continuous Gas Lift

- In the continuous gas lift, only one valve will be accomplishing the gas injection work and that this valve should be as deep as possible as per the available normal gas injection pressure, called Operating Valve.
- The valves above it are used to unload the well to initiate the flow from the reservoir, called Unloading Valves, and are closed once the gas injection begins through the operating valve
- Gas injection is done at a slow rate through GLV port sizes of 3⁄16", ¼" and 5⁄16".

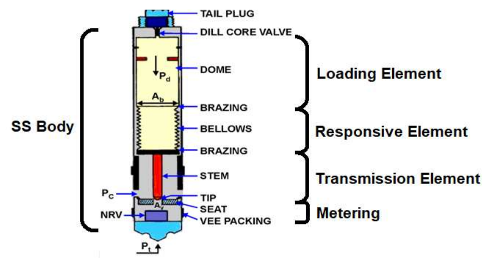


## Intermittent Gas Lift

- In an intermittent gas lift, a sufficient volume of gas is injected at the available injection pressure as quickly as possible into the tubing under a liquid column and then gas injection is stopped.
- The gas expands and in this process, it displaces oil onto the surface.
- In this system, an idle period is provided when no gas injection takes place and well is allowed to build up the level of the liquid.
- Intermittent gas lift is applicable for the wells that have the following characteristics:
- High PI, low bottom hole pressure
- Low PI, low bottom hole pressure
- Low PI, high bottom hole pressure

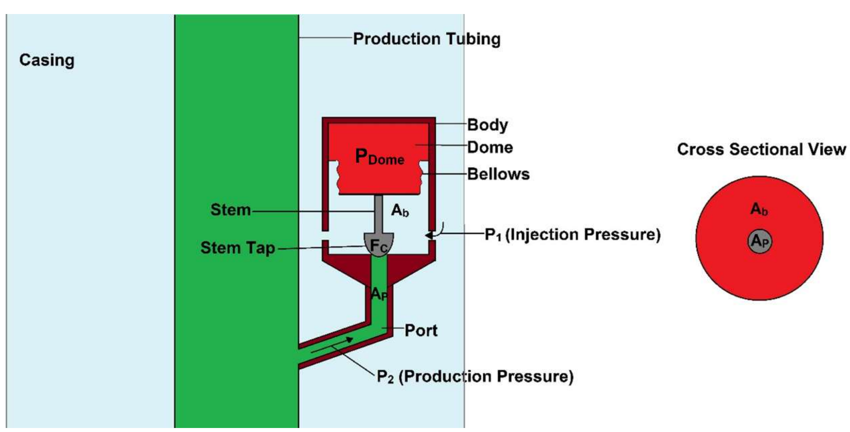
# Gas Lift Valve

- It is the heart of the gas lift system through which gas is injected.
- The bellow-operated nitrogen pressure loaded gas lift valve is the most common type of gas lift valve being used by oil industries.
- A gas lift valve has five basic components viz.
  - Body
  - Loading Element
  - Responsive Element
  - Transmission Element
  - Metering Element






# Derivation of Injection Pressure for GLV



# Derivation of Injection Pressure for GLV

Let,

- $P_{inj}$ be the injection pressure
- $P_{dome}$ be the pressure inside the dome
- $P_{prod}$ be the production pressure
- $F_C$ be the closing force
- $A_b$ be the cross sectional area of the bellow
- $A_P$ be the port area

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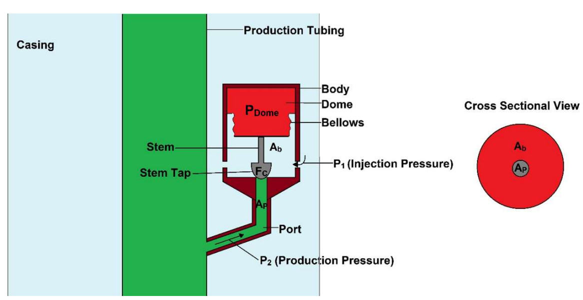
# Derivation of Injection Pressure for GLV

Applying force balance,

$$\text{Injection Pressure} + \text{Production Pressure} \geq \text{Dome Pressure}$$

$$[P_{inj} \times (A_b - A_P)] + [P_{prod} \times A_P] \geq F_C$$

$$[P_1 \times (A_b - A_P)] + (P_2 \times A_P) \geq A_b \times P_b$$

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## Derivation of Injection Pressure for GLV

$$\left[P_{inj} \times (A_b - A_P)\right] + \left(P_{prod} \times A_P\right) \geq A_b \times P_{dome}$$

$$\left[P_{inj} \times (A_b - A_P)\right] + \left(P_{prod} \times A_P\right) \geq A_b \times P_{dome}$$

$$\left[P_{inj} \times (A_b - A_P)\right] \geq (A_b \times P_{dome}) - \left(P_{prod} \times A_P\right)$$

$$P_{inj} \geq \frac{(A_b \times P_{dome}) - \left(P_{prod} \times A_P\right)}{(A_b - A_P)}$$

$$P_{inj} \geq \frac{\frac{(A_b \times P_{dome}) - \left(P_{prod} \times A_P\right)}{A_b}}{\frac{(A_b - A_P)}{A_b}}$$

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## Derivation of Injection Pressure for GLV

$$P_{inj} \geq \frac{\frac{(A_b \times P_{dome})}{A_b} - \frac{\left(P_{prod} \times A_P\right)}{A_b}}{1 - \frac{A_{Port}}{A_{bellow}}}$$

$$P_{inj} \geq \frac{P_{dome} - P_{prod}\left(\frac{A_P}{A_b}\right)}{1 - \frac{A_P}{A_b}}$$

$$P_{inj} \geq \frac{P_{dome} - P_{prod}M}{1 - M}$$

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# PETROLEUM PRODUCTION ENGINEERING

# MODULE-13

## INTRODUCTION TO ARTIFICIAL LIFT TECHNIQUES

## PROGRESSIVE CAVITY PUMP (PCP)

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## Progressive Cavity Pump (PCP)

- Progressing cavity pump is the youngest to join the family of different artificial lift modes.
- It's use in oil production started in early 1980s.
- Presently, it is mostly used in areas of:
  1. Heavy oil production
  2. De-watering of Coal Bed Methane (CBM) wells
  3. Oil production with sand cuts to 85%
  4. Mature water floods
  5. Visual or height sensitive areas

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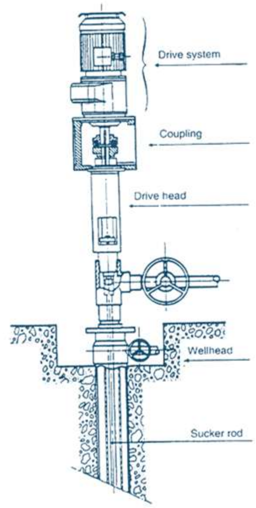
## Progressive Cavity Pump (PCP)






## Progressive Cavity Pump (PCP)

- It is a low capital cost, low operating cost, high efficiency mode of artificial lift. But, it's use is limited by:
  - i. Maximum pumping rate – around 250 $m^3$/d.
  - ii. Maximum depth of pump – around 2000 m.
  - iii. Maximum bottom hole temperature – around 150 °C.
- PCPs are being extensively used for heavy oil production in Canada, China, Venezula etc. where most of the world's heavy oil exists.
- Also being widely used in about thousand wells of Coal Bed Methane in USA for de-watering the wells.

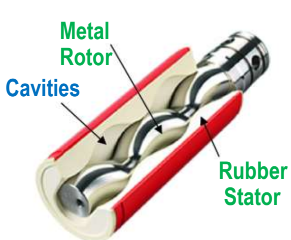
# Advantages

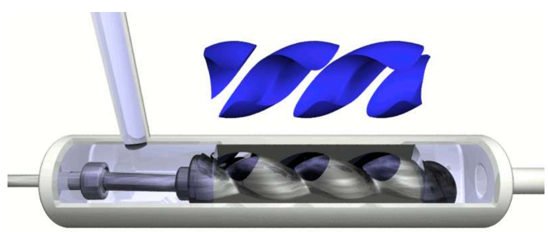
- Produce high viscous fluids, large solid concentration and moderate free gas
- No valves to clog or gas lock
- Ability to tolerate high percentages of free gas
- Low internal shear rates (limits fluid emulsification)
- Low capital and operating costs
- Low wellhead profile



# Operating Principles

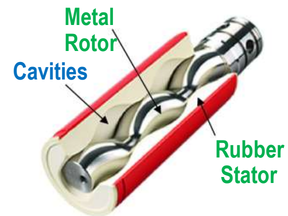
- A Progressing Cavity Pump essentially constitutes of two helical shafts, one inside the other and rotating around their longitudinal axis which are parallel but spaced between each other.

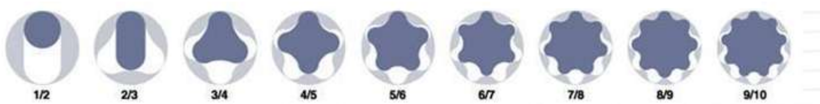


## Operating Principles

- The external gear has one more thread or lobe than the internal element.
- The internal member, known as rotor or bolt, is made of high strength steel and the outer member, called stator, formed from elastomer core permanently bonded to a steel tube.

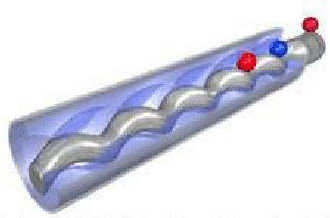




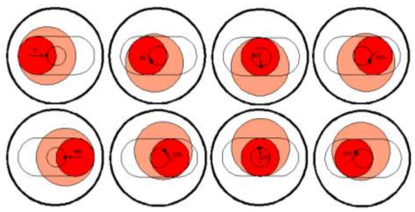

## Operating Principles

- For the simplest PCP geometry, the rotor is a simple helix.
- The geometry of assembly is such that it constitutes a series of identical, separate cavities.
- When the rotor is rotated inside the stator, these cavities move axially from one end of the stator to the other, from suction to discharge, creating pumping action.



## Operating Principles

- The internal element is set so that all the threads or teeth are constantly in contact with the external element.
- The helical pitches of the two elements are proportionate to each cross section with the number of teeth.
- This motion leads to the formation of "closed cavities", delimited by the rotor and the stator which move axially from suction to discharge.



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## Operating Principles

- A rotating positive displacement pump which is reversible and self-priming, without any check valve, providing a uniform flow rate, without any pulsation or jerk and capable of conveying very liquid to very pasty fluids, even when containing solids and gas.
- Two conditions are necessary to obtain "closed cavities":
  1. The rotor must have one tooth less than the stator and every tooth of the rotor must always be in contact with the inner surface of the stator.
  2. The rotor and the stator, as they have been defined above, constitute longitudinally two helical gears.

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## Limitations

- Limited depth capability of 8000 ft.
- High temperature degrades the elastomer beyond 150°C.
- Sensitivity to produced fluids as elastomer may swell or deteriorate when exposed to certain fluids.
- Low volumetric efficiencies when producing large amount of gas.
- Potential for tubing & rod coupling wear in directional and horizontal wells
- Requires constant fluid level above pump avoiding dry run which heats



# PETROLEUM PRODUCTION ENGINEERING

# MODULE-14

### INTRODUCTION TO ARTIFICIAL LIFT TECHNIQUES

### ELECTRIC SUBMERSIBLE PUMP (ESP)

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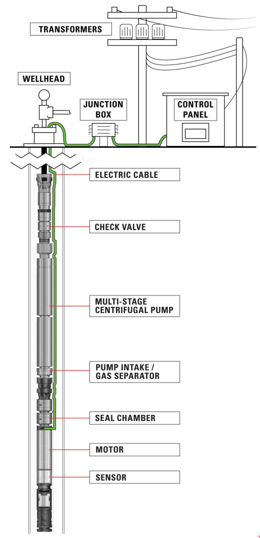
# Electric Submersible Pump (ESP)

- The electrical submersible pumping system is considered an effective and economical means of lifting large volumes of fluids from great depths under a variety of well conditions.
- The electrical submersible pump (ESP) is basically a high volume mode of lift system.
- 5 Distinct Components:
  - Pump Section
  - Separator Section
  - Seal Section
  - Subsurface Motor
  - Subsurface Cable

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# ESP-Sub-surface Motor

- The electrical submersible pump motor is of two-pole three-phase squirrel cage induction type one.
- These motors operate at a nominal speed of 3500 rpm on 60 Hz and 2915 rpm on 50 Hz cycle.
- These motors are filled with highly refined mineral oil to provide the necessary dielectric strength as well as a good thermal conductivity which prevent motor from getting overheated and thereby damaged.
- In general the motor, is attached to the bottom-most part of the assembly.

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## ESP-Separator Section

- The pump intake is connected in bolt-on-fashion to the lower side of the pump section of the electrical submersible pump and to the top of the protector.
- This provides a path for the fluid to enter into the pump.
- Very often, the straight intake section is replaced by the other forms of intake sections called gas separator for separating out the free gas from the liquid before the liquid enter the pump.
- The free gas is routed up through the annulus to ultimately get discharged in the flowline.



## ESP-Separator Section

- 2 Types available:
- The first one is the poor-boy type gas separator where the fluid bends 180 degrees and looses momentum releasing gases.
- The second category is the rotary type separator where the fluid enters the separator and is forced into a rotating centrifuge chamber by the action of an inducer.
- Once in the centrifuge, the fluid with the higher specific gravity is forced to the outer wall of the rotating chamber by centrifugal force, leaving the gas in near the center.

## ESP-Seal Section

- It protects the motor and it does the following functions:-
  - Allow for expansion and contraction of the dielectric oil contained in the rotor gap of the motor.
  - Equalize the casing annulus pressure with the internal dielectric motor fluid that helps to keep well fluid from leaking past the sealed joints of the motor.
  - Isolate the well fluid from the clean dielectric motor fluid as contamination of the motor insulation with well fluid can lead to early insulation failure.
  - Absorb the down thrust of the pump by a sliding thrust bearing



## ESP-Subsurface Motor

- Centrifugal pumps in a multi-stage fashion.
- Because of the ID of the casing, the diameter of the pump is very much restricted.
- The OD and the type of impeller design determines the rate of fluid production.

$$BHP = \frac{Q \times Head \times Specific\ Gravity}{Pump\ Efficiency}$$

- Whereas, the number of stages where each stage consists of one impeller and diffuser, are governed by the requirement of the head of fluid to be lifted to the surface against the given tubing pressure (H).

## ESP- Motor Shroud

- Motors can be damaged very quickly if they are not cooled.
- Sometimes, the liquid passing through the motor is not enough to cool the motor properly.
- In such cases, a motor shroud is used to divert/channel all the liquid pass the motor so that the liquid can flow very rapidly.



## ESP- Subsurface Cable

- Power is transmitted to the submersible motor by banding a specially constructed three phase electric power cable to the production tubing
- Cables are available in a wide range of conductor sizes, which permits efficient matching to motor requirements.
- The insulation used for these cables must be able to withstand wellbore temperatures, Pressure, and resist impregnation of well fluids.
- Designed in such a way that high voltage requirement is fulfilled, which means low current, that reduces the diameter of the conductors below 0.25 inch.

## Advantages of ESP

- High volume and depth capability
- Highly efficient
- Low maintenance
- Minor surface equipment need
- Good in deviated wells
- The system is easy to adapt to automation and can pump continuously and intermittently.
- For shallow wells, the investment is low



## Disadvantages of ESP

- Limited Adaptability to Major Changes in Reservoir
- Difficult to repair in the field.
- The build-up of scale deposits can interfere with the operation of submersible pumps.
- Also, the cost of electricity can also be very high, especially in remote areas where long distance supply connection is required.
- The system has limited flexibility under some producing conditions, and the entire system in the well must be pulled when a problem is encountered.

# PETROLEUM PRODUCTION ENGINEERING

# MODULE-15

## INTRODUCTION TO ARTIFICIAL LIFT TECHNIQUES

## HYDRAULIC LIFTS (JET PUMPS)

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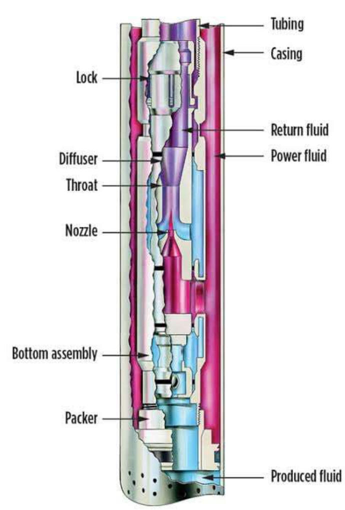
## Hydraulic Lifts (Jet Pumps)

- Jet pumps enable energy from a high pressure (HP) source to be used to boost the pressure of low pressure (LP) fluids.
- This is accompanied by momentum transfer of fluid through an ejector nozzle, throat and diffuser.
- The unit shown consists of a nozzle, a suction chamber, a mixing tube (throat) and a diffuser.



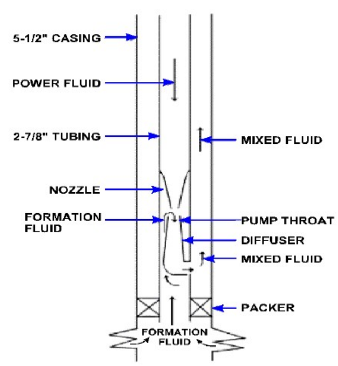
## Hydraulic Lifts (Jet Pumps)

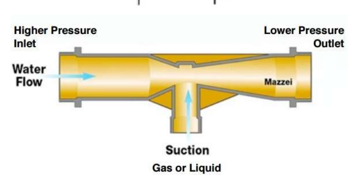
- High pressure fluid passes through the nozzle where part of its potential (pressure) energy is converted into kinetic energy (velocity).
- As a result, the pressure downstream of the nozzle drops.
- At this point, the surrounding well fluid, having comparatively higher fluid pressure and having access to the throat chamber will be sucked through production inlet chamber.
- The mixture passes through a mixing tube where transfer of momentum and energy takes place between well fluid and power fluids.



## Hydraulic Lifts (Jet Pumps)

- On passing from throat to diffuser, the mixture of well fluid and power fluid loses velocity and consequently acquires equivalent discharge pressure to a value sufficient enough to lift the total fluid to the surface.
- When the fluid passes through narrow/ constricted part of venturi tube it speeds up which is accompanied by decrease in fluid pressure.



## Advantages of Jet Pumps

- Positive displacement – strong drawdown.
- Double-acting high-volumetric efficiency.
- Good depth/volume capability – more than 15,000 ft.
- Operates in deviated wells.
- Multi-well production from single surface package.

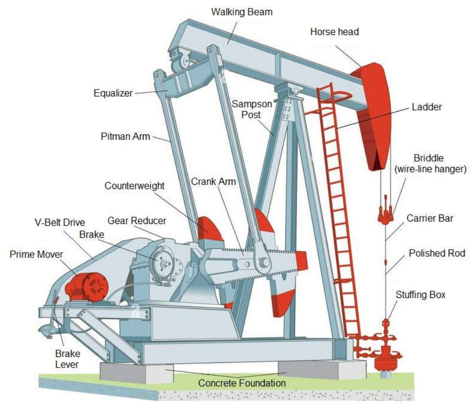


## Limitations of Jet Pumps

- Solids handling
- Requires specific bottom hole assemblies
- Medium volume potential (50 -1000 BPD)
- Require service facilities
- Gas handling limitations
- Requires high-pressure surface line

# Test Your Knowledge

- Illustrate any five (5) components of the surface unit of a Sucker Rod Pump used in the oil and Gas industry (shown below)




10 marks



# Test Your Knowledge

- Appraise on the pumping cycle of a sub-surface pump of a sucker rod pump used in the oil and gas industry 10 marks
- Illustrate on the Gas Lift System and the types of Gas Lift used in the oil and gas industry with a neat diagram. 15 marks
- Derive the equation for Gas Lift injection pressure with a neat diagram labelling the components. 15 marks
- Discuss the operating principles of a progressive cavity pump with a diagram labelling its parts 15 Marks
- Illustrate on the Progressive Cavity Pump (PCP) with a diagram labelling its parts and discuss the advantages and limitations of PCP. 15 marks

# Test Your Knowledge

- **Appraise on the Electrical Submersible Pump (ESP) used in the oil and gas industry with a labelled diagram.** **15 marks**
- **Discuss the advantages and disadvantages of an Electrical Submersible Pump (ESP) used in the oil and gas industry.** **10 marks**
- **Illustrate on the working of Jet Pump with a diagram labelling its parts and discuss the advantages and limitations of Jet Pump.** **15 marks**

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